

00_READ_FIRST — K3 SWARM: SIDE24/q0 LOWER-RATE RIGOROUS REPAIR

Work order: K3_SWARM_MASTER_WORK_ORDER_—_SIDE24_q0_L1.txt (2026-08-05, sha256 recorded in CANONICAL_STATE.json). Lead coordinator: KIMI line. This directory is the campaign's deliverable root. All dependency references inside deliverables use relative paths only.

Controlling facts established at intake (2026-08-05)

1. **The K3 blocking defect is CONFIRMED, numerically and symbolically.** The WP “certified upper bound” used the factor $p_{\text{grad}} \cdot \sqrt{E[\det(H)^2 \mid \text{grad}=0]} \cdot \min(P_{\text{type}}, P_{\text{window}})$. The Cauchy–Schwarz chain gives $\dots \cdot \sqrt{P(\text{type} \cap \text{window})} \leq \dots \cdot \sqrt{\min(P_{\text{type}}, P_{\text{window}})}$. Since $p \sim \sqrt{p}$ on $[0,1]$, the missing square root makes the reported quantity SMALLER — not a valid generic upper bound. At the certified witness point $(-0.04, -0.58)$: missing-sqrt value $1.31310\text{e-}4$ vs CS-valid $2.30247\text{e-}2$ ($175.3\times$ inflation). Consequences (all downstream uses INVALID as upper bounds): the advertised $I_{\text{cs}} \sim 1.30\text{e-}5$; the WP rung “upper-bound” table; the fitted $5.5\text{e-}3\text{-}r^{1.6}$ envelope; the AO reliability coefficient's WP-ub branch; THM-023 v1.1's WP row and its ub-based constant display. NOT affected: LB-1's above-b saddle channel ($\rho_{\text{CS}}/\rho_{\text{sad}}$ carry the sqrt — valid); the two-sided quadrature ESTIMATE $I_{\text{true}} \sim 4.76\text{e-}6$ (an estimate, never a bound claim; its own certification is a separate open task); the conditional-law witness values; the $m(P)/m(Q)$ premise-falsification points; Task 4c's Kantorovich enclosures; DER-027a's variance channel (independent machinery).
2. **THM-023 v1.1 is a noncontrolling HOLD draft** (unsealed PLACEHOLDER-BODY-HASH, [PENDING] slots, an unfinished certificate path with a KeyError: verifier accesses tbl[‘0.005’] absent from its RUNGS). Its bytes are preserved unchanged in the research tree; nothing in this campaign consumes it.
3. **The 0.9144036 coefficient is a candidate measured/mixed-tier asymptotic anchor**, not a proved fixed theorem constant. The successor assembly must keep the four tiers distinct (proved constant / limiting coefficient / measured anchor / direct numerical estimate).
4. **The manuscript delivered 2026-08-05** (SIDE24_pre_peer_review) consumed the invalid bound in its §3.10.6/3.10.9/§6 (the $1.30399\text{e-}5$ “certified upper bound”, the $5.5\text{e-}3\text{-}r^{1.6}$ envelope, and the ub branch of the constant ledger). A corrected revision is owed after the WP adjudication; the invalid items are quarantined here as evidence pending W2/W3/W4 conclusions.
5. Corpus: 578 uploaded files + 20 research carriers inventoried byte-exact (INPUT_MANIFEST.sha256, FILE_LEDGER.tsv). No MISSING objects at intake. The orphaned pre-update THM body 40596829... remains declared UNRECOVERABLE (KIMI-DATA-028; successor citations re-point to ba3974c0...).
6. Estimand firewall: $p_r, q_r, q_r^{\text{adj}}$, elder-rule pairing, and the ratified upper theorem are distinct; every typed-to-adjacent transfer retains the $_r$ channel. The ratified upper chain (AO48-OPR-045) is outside this campaign. This work changes no Boolean/operator status.

Swarm map (workstream → owner → deliverable)

- W1 corpus/provenance: lead (this directory's intake artifacts).
- W2 symbolic WP/Kac–Rice auditor: independent agent (freezes derivation before seeing numerics).
- W3 rigorous WP numerics: independent agent (interval enclosures).
- W4 independent WP implementation: independent agent (different representation; freeze-then-compare).

- W5 C030/C031 forensic: agent (provenance of $3e-15$ and $0.213r^3$).
- W6 uniform-in-r analyst: agent (scaled asymptotic law or interval coverage).
- W7 -LOC: existing DER-026 agent (context retained).
- W8 Λ -side: existing DER-027b agent (context retained; full-grid transcripts running).
- W9 DER-027a/027c scope: existing DER-027a/027c agents.
- W10 dependency-minimization: agent (event-inclusion recomposition; what each channel must really prove).
- W11 literature/methods: agent (primary sources).
- W12 independent adversarial reviewer: dispatched after W2/W3/W4 freeze.
- W13 reproducibility/export: dispatched with the assembly.

Phase discipline

W2, W3, W4 must NOT see each other's conclusions until each files a frozen report (hash-bound). The lead routes cross-context only after the freeze receipts exist. No theorem successor is minted until gates G0–G12 pass; a conspicuously marked noncontrolling shell is prepared first.

Corrections (red-team findings, 2026-08-05)

- **B2 (Isserlis display, lead).** Fact 1's canonical witness absolutes were computed with a fourth-moment display that drops mean terms (overcount exactly $2(\det \nu \nu^T)^2$; inflation $\sqrt{1.34488}$). Corrected: defective form $1.31310e-4 \rightarrow 1.13229e-4$; CS-valid $2.30247e-2 \rightarrow 1.98542e-2$; the defect ratio 175.35 and the D1 quarantine are unchanged; direction conservative. See LEAD_SYMBOLIC_WP_CORRECTION_ADDENDUM.md.
- **B1 (W6 modulus).** $E_{WP} = 4.2 * r^{\{3/2\}}$ is invalid for $r \lesssim 0.003$ (near-cluster CS singularity artifact); $E_{WP}(0.05)$ survives; W6 is repairing via zone split. WP-min is therefore NOT yet delivered (G6 open).
- **B3 (F1 quantification).** W6's premise F1 was mis-quantified (worst gap $\sup 4.236 > 4.2$ under the stated premise; 4.129 under measured slope 0.20). Part of the W6 repair.
- **B4 (gamma-LOC scope).** W7's retirement holds for Forms B/C/D; the exit node of Form A still consumes P-NMZ-gamma. Re-scoped accordingly.